

RESHMA RAGHAVAN

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PROFESSIONAL EXPERIENCE

Geospatial Web Developer Intern

Aquaveo, LLC

January 2026 - present

Provo, UT, USA

- Develop and maintain geospatial web applications using Python (Tethys Platform) and JavaScript (React.js)
- Implement web-mapping features integrating ArcGIS services, ESRI JSON, GeoJSON, and GeoTIFF datasets
- Assist with building data models and workflows to support processing and visualization of geospatial data
- Create interactive maps and charts to improve accessibility and understanding of spatial and hydrologic data
- Support development of flood inundation mapping tools used in academic and research environments

Software Engineer

Scientific Computing and Imaging Institute

September 2024 - September 2025

Salt Lake City, UT, USA

- Engineered full-stack, cloud-enabled, ML/AI solutions for NSF-funded National Data Platform and Democratizing Data initiatives
- Implemented LLM-based pipelines to assist with data validation and improve consistency across datasets
- Built and updated user interfaces using Next.js to improve usability and application performance
- Containerized Node.js services with Docker Compose to support development and deployment workflows
- Assisted with CI/CD pipeline configuration using GitHub Actions to support automated deployments
- Implemented Keycloak and OAuth2 authentication for secure access across internal applications
- Supported deployment of services on AWS EC2 and EKS environments

Research Assistant (Software Developer)

University of Utah

June 2024 - September 2024

Salt Lake City, UT, USA

- Implemented a new state management system in Next.js, improving legacy system integration by 60% and reducing re-renders
- Streamlined ML workflows with Docker, reducing consumption by 15% through optimized configurations across operating systems
- Automated data pre-processing and DVC pipelines, speeding experiment reproduction and cutting manual data tasks

Jr. Machine Learning & Object Detection Engineer

PassiveLogic, Inc.

February 2024 - June 2024

Holladay, UT, USA

- Automated data preparation and model training on Google Cloud (GCP) using Vertex AI to implement a YOLO-driven object detection pipeline for PassiveLogic's iOS application, enabling real-time identification of 10 distinct object types with an average precision of 88% on mobile devices

TECHNICAL PROJECTS

Recipe Sharing Website

July 2024

- Developed a full-stack recipe platform using **React.js** and **Django**, leveraging strong web development fundamentals
- Deployed the React frontend (**JavaScript**) as static files on **AWS S3** for scalable hosting
- Launched the Django backend (**Python**) on **AWS EC2**, integrating with an **SQLite** database for recipe storage and retrieval

Solar Water Heater Simulation

August 2023

- Built a web app simulating solar water heater performance, visualizing temperature change and system efficiency
- Designed the **frontend** with **raw HTML**, **CSS**, and **JavaScript**, incorporating **Chart.js** for **real-time line graph visualization**
- Implemented a **Node.js backend** following the **MVC architecture** and software architecture best practices, with **unit testing using Jest** to ensure reliability and code quality

EDUCATION

Master of Software Development

Kahlert School of Computing

University of Utah

2022 - 2023

Salt Lake City, Utah

GPA: 3.9/4.0

Master of Business Administration (Communication Management)

Symbiosis Institute of Media and Communication

2013 - 2015

Bengaluru, Karnataka, India

Bachelor of Science in Visual Communication

M.O.P Vaishnav College for Women, Chennai (Autonomous)

2009 - 2012

Chennai, Tamil Nadu, India